Problem Solving

Answer key

Conventions used throughout

• **Forces:** F_g = gravity (mg), F_N = normal force, T = tension. F_{1y} means the y component of F_1.

• **Signs:** in Question 1, +x is right and +y is up, with signed components. In the pulley questions (4 and 5), each mass gets its own + direction, the way it moves (as in 2.4), and forces are written as sizes with + or − in the equation.

• **Pulleys (Questions 4 and 5):** try both choices of +x along the string; the case with a positive a is the real motion. Final answers give the scalar with its sign, then the vector with [positive x]. Each unknown is isolated as a formula before the numbers go in.

• **Kinematics:** v_1 = initial velocity, v_2 = final velocity, Δd = displacement.

• g = 9.81 m/s^2; no friction or air resistance unless the question gives it. Every step keeps full precision; only final answers are rounded.

QUESTION 1

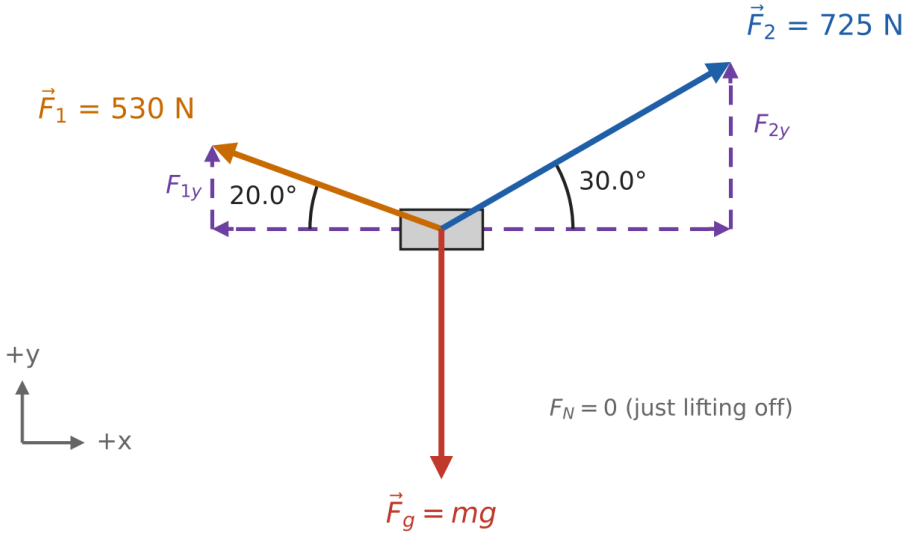
Lifting a box with two ropes

Complete the following additional problems for practice.

1. Two people pull on ropes as illustrated below:

Find the mass of the object if the forces applied cause the object to just lift off the ground.

FBD of the object (+x right, +y up)



Step 1 – What “just lift off” means

As the pulls get stronger, the ground pushes up less and less. At the moment the object lifts off, the ground stops pushing: $F_N = 0$, and the object is not yet speeding up vertically, so $a_y = 0$. Only the y direction matters for the mass.

Step 2 – The y components of the two pulls

Each y part is opposite its angle above the horizontal, so it uses sine. Both point up (+):

$$F_{1y} = (530)\sin 20.0° = 181.27 \text{ N}$$

$$F_{2y} = (725)\sin 30.0° = 362.50 \text{ N}$$

Step 3 – Balance the y direction

Formula: $\Sigma F_y = ma_y$

Put in the forces, $F_N = 0$ and $a_y = 0$: $F_{1y} + F_{2y} - mg = 0$

Add mg to both sides: $mg = F_{1y} + F_{2y}$

Divide both sides by g: $m = \dfrac{F_1 \sin 20.0° + F_2 \sin 30.0°}{g}$

$$m = \frac{(530)\sin 20.0° + (725)\sin 30.0°}{9.81} = \frac{181.27 + 362.50}{9.81}$$

$$m = \frac{543.77}{9.81} = 55.43 \text{ kg}$$

The x parts (725 cos 30.0° = 627.87 N right, 530 cos 20.0° = 498.04 N left) don’t balance, so the object also slides to the right, but that doesn’t change the mass.

FINAL ANSWER

m = 55.4 kg

QUESTION 2

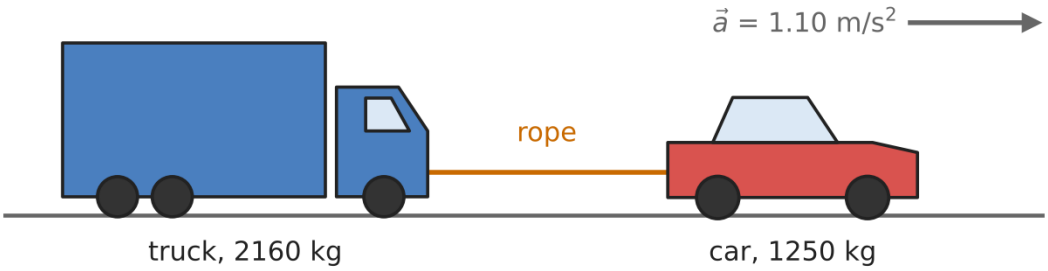
Car towing a truck

2. A 1250 kg car is towing a 2160 kg truck using a rope to connect the two vehicles. The acceleration after being stopped at a red light is 1.10 m/s^2 forward once the light turns green.

a) Find the force needed to accelerate the connected vehicles.

b) What minimum strength needed in the rope connecting the two vehicles?

Picture the situation



The car pulls the truck forward through the rope. No friction is given, so it is ignored.

(a) Find the force needed to accelerate the connected vehicles.

Treat the car and truck as one system. The rope is inside it and cancels, so the forward force on the system has to accelerate both masses:

Formula: $F_{net} = ma$

$$F_{net} = (1250\text{ kg} + 2160\text{ kg})(1.10\text{ m/s}^2) = (3410)(1.10) = 3751\text{ N}$$

(b) What minimum strength needed in the rope connecting the two vehicles?

The rope is the only force pulling the truck forward, so the rope's tension is the truck's net force:

$$T = m_{truck}\, a = (2160 \text{ kg})(1.10 \text{ m/s}^2) = 2376 \text{ N}$$

The rope must be able to hold at least this tension without breaking.

FINAL ANSWER

(a) $\vec{F}$ = 3.75 × 10^3 N [forward]

(b) The rope must hold at least 2.38 × 10^3 N

QUESTION 3

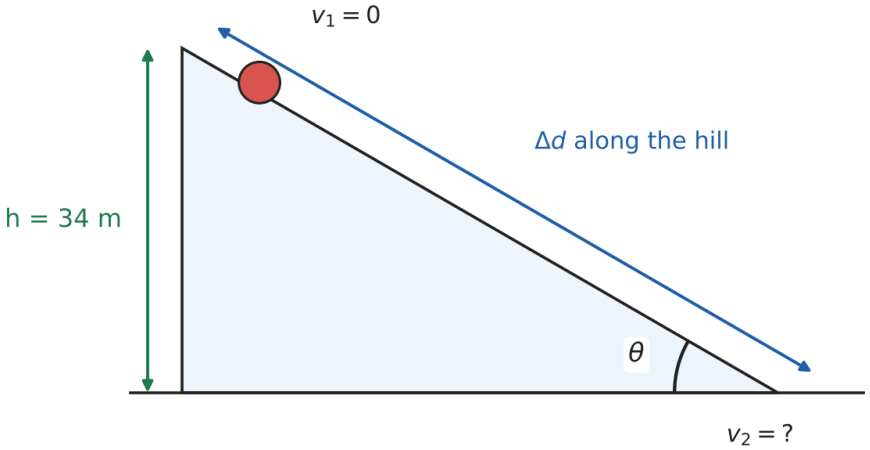
Toboggan down a hill

3. A boy on a toboggan is sliding down a hill that makes an angle θ to the horizontal. The height of the hill is 34 m.

a) What will be the speed of the toboggan at the bottom of the hill?

b) Explain why the answer to part (a) does not require the angle of the hill or the mass of the toboggan and boy.

Picture the situation



No friction and no starting speed are given, so the toboggan starts from rest on a frictionless hill.

(a) What will be the speed of the toboggan at the bottom of the hill?

Step 1 – Acceleration down the hill

Take +x down the hill. The only force along the hill is the part of gravity along it, mg sin θ:

Formula: $\Sigma F_x = ma_x$

Put in $\Sigma F_x = mg\sin\theta$: $mg\sin\theta = ma_x$

Divide both sides by m: $a_x = g\sin\theta$

Step 2 – Length of the hill

The height h is opposite θ and the hill's length Δd is the hypotenuse:

Formula: $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}} = \dfrac{h}{\Delta d}$

Multiply both sides by Δd and divide by $\sin\theta$: $\Delta d = \dfrac{h}{\sin\theta}$

Step 3 – Final speed

Use the kinematics equation with no time in it:

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_1 = 0,\ a = g\sin\theta,\ \Delta d = \dfrac{h}{\sin\theta}$: $v_2^2 = 2(g\sin\theta)\left(\dfrac{h}{\sin\theta}\right)$

The $\sin\theta$ cancels: $v_2^2 = 2gh$

Square root both sides: $v_2 = \sqrt{2gh} = \sqrt{2(9.81)(34)} = 25.83$ m/s

(b) Explain why the answer to part (a) does not require the angle of the hill or the mass of the toboggan and boy.

Mass: gravity along the hill is mg sin θ, and Newton's second law divides by m, so the mass cancels (a = g sin θ). A heavier toboggan gets a bigger force but has more mass to move.

Angle: a steeper hill gives a bigger acceleration (g sin θ), but the hill is shorter (h ÷ sin θ). The two effects cancel exactly in $2a\Delta d$, leaving $v_2^2 = 2gh$. Only the height matters.

FINAL ANSWER

(a) $v_2 = 26$ m/s

(b) The mass cancels in a = g sin θ, and sin θ cancels between a and Δd, leaving v = √(2gh)

QUESTION 4

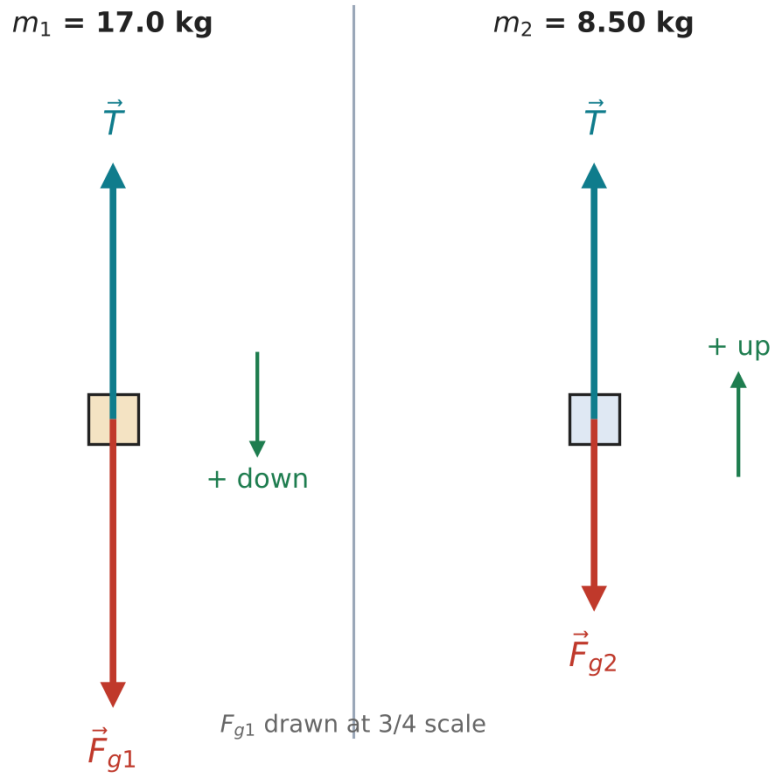
Atwood machine

4. An 8.50 kg mass and a 17.0 kg mass are connected by a string over a pulley as shown in the figure below:

a) Calculate the acceleration of the system.

b) Calculate the tension in the string that connects the two masses.

FBDs for Case 1 (m_1 = 17.0 kg: +x down; m_2 = 8.50 kg: +x up)



(a) Calculate the acceleration of the system.

We don’t know the direction until we calculate, so try both choices of +x along the string. As one system the tension cancels, and only the two weights act along the string.

Case 1 – +x is m_1 down, m_2 up

m_1g points along +x and m_2g against it. Solve for a in letters first:

Formula: $F_{net} = (m_1 + m_2)\,a$

Put in the forces: $m_1g - m_2g = (m_1 + m_2)\,a$

Divide: $a = \dfrac{(m_1 - m_2)\,g}{m_1 + m_2}$

$$a = \frac{(17.0 - 8.50)(9.81)}{17.0 + 8.50} = \frac{83.385}{25.5} = +3.270 \text{ m/s}^2$$

Positive, so both masses accelerate in this +x direction.

Case 2 – +x is m_1 up, m_2 down

$$m_2g - m_1g = (m_1 + m_2)\,a \quad \Rightarrow \quad a = \frac{(m_2 - m_1)\,g}{m_1 + m_2} = -3.270 \text{ m/s}^2$$

Negative, so both masses accelerate opposite to this +x, which is the Case 1 direction.

(b) Calculate the tension in the string that connects the two masses.

Look at m_2 by itself (Case 1, +x up). Isolate T in letters, then put in the numbers:

Formula: $F_{net} = ma$

Mass 2: $T - m_2g = m_2a$

Add m_2g and factor: $T = m_2(g + a)$

$T = (8.50)(9.81 + 3.270) = 111.18$ N

FINAL ANSWER

(a) a = +3.27 m/s^2 (Case 1)

$\vec{a}$ = 3.27 m/s^2 [positive x], the same for both masses

(b) T = 111 N

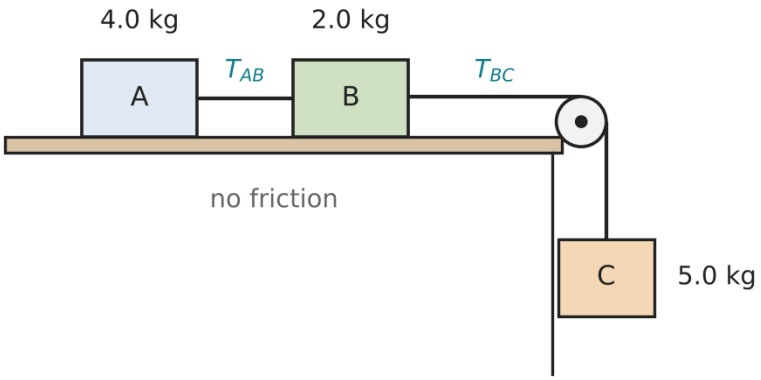
Check: mass 1: T = m_1(g − a) = (17.0)(9.81 − 3.270) = 111.18 N.

QUESTION 5

Two blocks on a table and one hanging

5. Use the following diagram to determine the tension force between mass A and B. A = 4.0 kg, B = 2.0 kg and C = 5.0 kg. There is no friction.

Picture the situation



A and B are on a frictionless table; C hangs from the string over the pulley.
On the table, F_N and F_g cancel for A and B, and there is no friction. As one system, the only outside force along the string is C's weight.

Case 1 – +x is C down (A and B toward the pulley)

$m_C g$ points along +x:

Formula: $F_{net} = (m_A + m_B + m_C)\,a$

Put in the force and divide: $a = \dfrac{m_C g}{m_A + m_B + m_C}$

$$a = \frac{(5.0)(9.81)}{4.0 + 2.0 + 5.0} = +4.459 \text{ m/s}^2$$

Positive, so all three blocks accelerate in this +x direction. (Case 2, +x with C up, gives a = −4.459 m/s^2: the same motion, opposite to that +x.)

Tension between A and B

T_{AB} is the only sideways force on A. Isolate it in letters, then put in the numbers:

Mass A: $T_{AB} = m_A a$

Put in a : $T_{AB} = \dfrac{m_A m_C g}{m_A + m_B + m_C}$

$$T_{AB} = \frac{(4.0)(5.0)(9.81)}{4.0 + 2.0 + 5.0} = 17.84 \text{ N}$$

FINAL ANSWER

T_{AB} = 18 N